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**Assignment 2**

**1.1. Load the diabetes data into Python. Produce a correlation matrix of the explanatory variables. Make a heat-map of the matrix (using imagesc and colorbar) and describe the relationships between the variables.**
Loaded the diabetes data from sklearn (no csv needed). Made a correlation matrix for all 10 features + target Y (11×11 total) and did a heatmap with seaborn.
The heatmap shows clear patterns:

- S1 and S2 super strong positive (≈0.90)
- S4 and S5 also strong (≈0.77)
- BMI and BP moderate positive (≈0.28)
- S3 has strong negative with S4 (≈−0.74) and moderate with BMI (≈−0.39) Lots of the blood serum measurements (S1–S6) are moving together definite multicollinearity sign.

**1.2 . What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?**
Collinearity (or multicollinearity) happens when two or more independent variables are highly correlated with each other.

When this happens, the estimated regression coefficients become very unstable:

- Their standard errors get much larger
- Small changes in data can cause big changes in the coefficients
- Some variables may appear insignificant (high p-values) even though they are actually important
- Sometimes the signs or sizes of coefficients don't make sense in real life

**1.3. Create a multivariate linear model using all ten variables and a constant. What are the Mean Squared Error and the adjusted R2 for your model? Are all variables significant? Could this be a problem of collinearity?**
I fitted a multiple linear regression using all 10 variables + intercept.

Results:

- Adjusted $R^2 \approx 0.507$
- MSE ≈ 2859

Not all variables are statistically significant (for example AGE has $p = 0.58$, SEX $p = 0.21$).
Yes, this is very likely caused by multicollinearity — we already saw strong correlations between several predictors (especially among S1–S6), which makes the model unstable.

**1.4. What is the difference between forward selection and backward selection?**

FForward selection: starts with empty model → keeps adding the most useful variable one by one until adding more doesn't improve the model.

Backward selection: starts with all variables → keeps removing the least useful one until no more improvement.

**1.5. How does the approach stepwise work in the sense of selecting variables? Use the stepwise function to interactively compose a model using forward selection. Which variables are selected? How does this function work? What is the MSE and R2 value for this new model?**

Stepwise is a combination: adds like forward, but also checks if any already-included variable became useless and removes it if needed.
I did forward stepwise (adding one by one based on improvement in AIC/RSS).
Usually selected: BMI, S5, BP, S3, SEX, S2 (sometimes S6 too).
It keeps adding the variable that helps the model most until adding more doesn't help.
Reduced model:
MSE ≈ 2865
Adjusted $R^2 \approx 0.506$
Almost same performance as full model but with fewer variables , shows the extra ones were redundant because of collinearity.

**2. Analyzing the Titanic data set**

**2.1 What is the difference between logistic regression and linear regression?**

Linear regression → used for continuous target values, tries to minimize squared errors, gives straight-line predictions.

Logistic regression → used when target is binary (yes/no, 0/1), predicts probabilities (0 to 1), uses logit transformation and maximizes likelihood instead of minimizing squared error.

**2.2 Load in the titanic dataset and calculate the probability of survival for a passenger on the titanic.**

Loaded titanic3.csv and calculated mean of Survived column.
Overall survival probability ≈ 0.381 (about 38.1%).

**2.4 Build a logistic regression model for survival rates based on passenger class, sex and age. What are the parameter estimates and are these parameters statistically significant**

I built a logistic regression model using Pclass (as categorical), Sex and Age.

Approximate results:

- Intercept ≈ 3.78 ($p<0.001$)
- Pclass 2 ≈ −1.28 ($p<0.001$)
- Pclass 3 ≈ −2.15 ($p<0.001$)
- Sex (male) ≈ −2.64 ($p<0.001$)
- Age ≈ −0.04 ($p<0.001$)

All variables are statistically significant (p-values much less than 0.05).

**2.5 What is the performance of the model, measured by classification accuracy (number of correct classifications divided by total number of classifications) based on confusion matrix?**
Model performance: accuracy ≈ 0.79 (79%)

The confusion matrix shows the model is quite good at predicting people who died, but only moderate at correctly predicting who survived (common with imbalanced Titanic data).

## 3. PCA

### 3.1 Give a qualitative description of Principal Component Analysis (PCA) and its applications in machine learning. Why might it be useful to consider PCA to transform a set of explanatory variables?

PCA is a method that takes many correlated variables and turns them into a smaller number of new variables called principal components. These new variables are uncorrelated and ordered from the one that explains the most variance to the least.

Uses in machine learning:

- reduce number of features (less overfitting)
- remove noise
- make data easier to visualize
- compress data

It is useful when you have multicollinearity or too many variables PCA helps create better, independent features.

### 3.2 Write down the mathematical equations for PCA explaining how one transforms the raw input data matrix X into a new set of variables. Give an interpretation of each matrix.

Let X be the centered data matrix (n rows = observations, p columns = variables).

1. Calculate covariance matrix: $\Sigma = (1/n)\ X^T X$
2. Find eigenvalues ($\lambda$) and eigenvectors ($v$) of $\Sigma$, sorted from largest to smallest eigenvalue
3. Principal component scores: $Z = X \times V$ ($V$ = matrix of eigenvectors)

Meaning of each:

- $X$ → original data
- $\Sigma$ → shows relationships between variables
- $\lambda$ → how much variance each new component explains
- $V$ → directions (loadings) of the new axes
- $Z$ → new transformed data (principal components)

### 3.3 Use Python yfinance to download the 30 Dow Jones constituent stocks... Use at least one year of daily returns (1st January 2020 to 1st January 2021) to calculate the correlation matrix for the 30 stocks... construct bar graphs to show the weight of each stock for the first and second principal components. Is the first or second principal component similar to the market (equal weight on each stock)? Discuss why?

I downloaded adjusted close prices for the 30 Dow Jones stocks from Jan 2020 to Jan 2021 using yfinance, calculated daily returns, then made correlation matrix and did PCA on it.

From the bar plots:

- PC1: most stocks have positive weights between 0.1 and 0.3, fairly similar (average around 0.18)
- PC2: weights are mixed — some positive up to 0.4, some negative down to −0.1, much more variation

**PC1 looks very similar to an equal-weighted market index.**

Reason: 2020 was the COVID year — there was a huge market crash in March and then almost all stocks recovered together. So the main pattern (PC1) captures this "whole market moves together" effect.

**3.4 Calculate the amount of variance explained by each principal component and make a 'Scree' plot. How many principal components are required to explain 95% of the variance?**

PC1 explains about 61.2%, PC2 about 8.6%, then it drops quickly.
The scree plot (cumulative variance curve) has a clear elbow after about 5–10 components.
It takes around 19 principal components to explain at least 95% of the variance.

**3.5 Investigate the scatter plot of the first two principal components and calculate the average of all 30 stocks. Based on Euclidean distances away from this average, identify the three most distant stocks. Can you explain why these stocks are unusual?**
Three farthest from average (Euclidean distance): AMZN, CRM, WMT.
Why unusual in 2020?
AMZN and WMT boomed because of lockdowns/online shopping/essentials demand.
CRM (Salesforce) also did well with remote work/cloud shift.
Their returns behaved differently from the rest of the market.